Homework Set C.2

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Problem 2

Let $u, v, w \in \mathbb{R}^3$ with components $u = (u_1, u_2, u_3), v = (v_1, v_2, v_3), w = (w_1, w_2, w_3)$.

$$(u+v) \times w = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_1 + v_1 & u_2 + v_2 & u_3 + v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = \begin{cases} \hat{i}\left((u_2 + v_2)\, w_3 - (u_3 + v_3)\, w_2\right) - \\ \hat{j}\left((u_1 + v_1)\, w_3 - (u_3 + v_3)\, w_1\right) + \\ \hat{k}\left((u_1 + v_1)\, w_2 - (u_2 + v_2)\, w_1\right) \end{cases}$$

$$= \begin{cases} \hat{i}\left(u_2 w_3 + v_2 w_3 - u_3 w_2 - v_3 w_2\right) - \\ \hat{j}\left(u_1 w_3 + v_1 w_3 - u_3 w_1 - v_3 w_1\right) + \\ \hat{k}\left(u_1 w_2 + v_1 w_2 - u_2 w_1 - v_2 w_1\right) \end{cases}$$

$$u \times w + v \times w = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_1 & u_2 & u_3 \\ w_1 & w_2 & w_3 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = \left\{ \begin{matrix} \hat{i}\left(u_2 w_3 - u_3 w_2\right) - \\ \hat{j}\left(u_1 w_3 - u_3 w_1\right) + \\ \hat{k}\left(u_1 w_2 - u_2 w_1\right) \end{matrix} \right\} + \left\{ \begin{matrix} \hat{i}\left(v_2 w_3 - v_3 w_2\right) - \\ \hat{j}\left(v_1 w_3 - v_3 w_1\right) + \\ \hat{k}\left(v_1 w_2 - v_2 w_1\right) \end{matrix} \right\}$$

$$= \left\{ \begin{matrix} \hat{i}\left(u_2 w_3 + v_2 w_3 - u_3 w_2 - v_3 w_2\right) - \\ \hat{j}\left(u_1 w_3 + v_1 w_3 - u_3 w_1 - v_3 w_1\right) + \\ \hat{k}\left(u_1 w_2 + v_1 w_2 - u_2 w_1 - v_2 w_1\right) \end{matrix} \right\} = (u+v)w$$

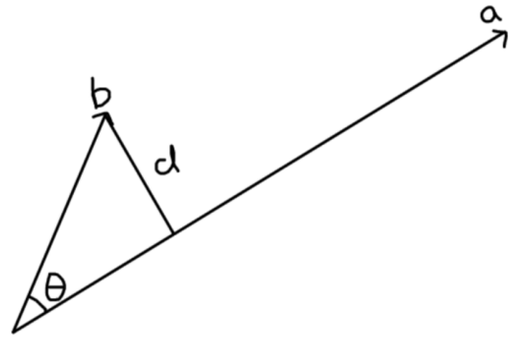
Problem 3

Let $a, b \in \mathbb{R}^3$. Notice that $a \times (-b) = (-a) \times b = -(a \times b)$.

$$\begin{aligned} (a-b) \times (a+b) &= (a + (-b)) \times (a+b) = a \times (a+b) + (-b) \times (a+b) \\ &= a \times a + a \times b + (-b) \times a + (-b) \times b = 0 + a \times b - (b \times a) - (b \times b) \\ &= 0 + a \times b - (-(a \times b)) - 0 = a \times b + a \times b = \boxed{2(a \times b)} \end{aligned}$$

Problem 4

Let θ be the angle between the vectors a and b.



$$\sin\theta = \frac{d}{\|b\|} \implies d = \|b\| \sin\theta = \frac{\|a\|\|b\|\sin\theta}{\|a\|} = \boxed{\frac{\|a \times b\|}{\|a\|}}$$